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Physical top soil properties in pure stands of Norway spruce (*Picea abies*) and mixed species stands in Austria

Torsten W. Berger^{*}, Herbert Hager

Institute of Forest Ecology, Univ. f. Bodenkultur, Peter Jordan-Straße 82, A-1190 Vienna, Austria

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Abstract

This study was done to evaluate the impact of pure Norway spruce stands on physical soil properties of top soil in comparison to mixed species stands on comparable sites. It was hypothesized that the flat root system of spruce causes soil compaction, which would have a negative impact on the soil aeration and hydraulic properties and consequently on seedbed quality, as well as early tree growth and seedling establishment. Hence, this topic is important for forest restoration, especially converting secondary pure spruce stands to mixed species stands. Forty-eight sites (24 pairs pure spruce stand/mixed species stand) of different stand development stages (mature stage, pole stage) were selected on two different bedrock materials (Molasse, Flysch). Undisturbed soil cores were taken from 0–4 and 4–8 cm soil depth and the following soil parameters were determined: (total) bulk density, remaining fine soil bulk density, dry masses of coarse fragments, roots and forest floor, organic carbon content, total soil pore volume as well as macropore volumes (after free drainage for 24 h and after water desorption at 10 kPa). The remaining fine soil bulk density was a useful parameter for characterization of the state of compactness. Pure spruce stands caused a lower bulk density of the upper mineral soil due to lifting and loosening of the soil above the root system. Results of calculated macropore volumes after water desorption at 10 kPa were exactly conform with those obtained for the remaining fine soil bulk density, indicating significant differences for the grouping variables bedrock material and species composition. It is concluded from this study that changes of soil physical properties of the upper mineral soil (0–8 cm soil depth) by Norway spruce will not reduce germination and growth of mixed species trees. However, chemical and nutritional changes were not subject of this study, which are expected to limit the success of forest restoration.

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1. Introduction

Picea abies is the dominant tree species in Austria (FBVA, 1998). While pure spruce stands are the natural vegetation in the subalpine region, monospecific stands of Norway spruce below elevations of

700 m must be considered manmade (Mayer, 1974). These secondary pure spruce stands are prevalent in many parts of the country, because high productivity coupled with good timber prices are tempting for the short term economic success. On the long term, however, forest site degradation (Schmidt-Vogt, 1986) and low stability of such stands which results among others in enhanced risks from windthrow (Rottmann, 1989) and pests (Schwerdtfeger, 1981) may cause lower profitability of secondary spruce stands than of natural mixed forests. That is why conversion of

^{*} Corresponding author. Tel.: +43-1-47654-4107; fax: +43-1-4797896.
E-mail addresses: tberger@woek.boku.ac.at, 104167.3573@compuserve.com (T.W. Berger)

secondary spruce stands to mixed species stands is a current issue in Europe.

Picea abies is commonly described as a tree with a flat root system with a much lower penetration energy than other coniferous trees (in decreasing order: *Abies alba*, *Pinus sylvestris*, *Pseudotsuga menziesii*, *Larix decidua*, *Picea abies*, *Pinus strobus*) and most deciduous trees (Rottmann, 1989; Korotaev, 1992). Hence, natural mixed forests of *Fagus sylvatica*, *Abies alba* and *Picea abies* root to deep mineral soil horizons, affecting the nutrient storage and turnover of the whole soil profile and not only of the upper mineral soil horizons as pure spruce stands do. Especially waterlogged horizons prevent deeper rooting of spruce and contribute to a plate-shaped root system (Xu et al., 1997).

Drexhage (1994) and Lee (1998) point out that classifying the root system of spruce as a flat system only is not correct, because it is very flexible, depending on the ambient soil conditions. These authors detected roots down to 140 cm soil depth at very favourable soil conditions: deep soil, no coarse fragments, no waterlogging, low bulk density, high pH and high base saturation. However, in all cases the dry mass of vertical coarse roots was less than 28% of the total mass of coarse roots, indicating that most of the root mass is spread horizontally in the upper mineral soil horizons and the organic layer. Many other authors conclude that waterlogging (Zoth and Block, 1992) and soil chemical conditions (Weiss and Agerer, 1986; Rastin and Ulrich, 1990) are important in the shallow rooting of spruce.

The impact of Norway spruce stands in comparison to natural mixed species stands on the soil structure is documented in several publications (Wiedemann, 1923; Krauss et al., 1939; Hauff et al., 1950; Richard, 1953; Schröder, 1954; Schlenker et al., 1969; Miehlich, 1970). These authors were specially interested in studying spruce on pseudogley, a soil type characterized by a seasonal waterlogging. The results of these studies are contradictory: Schlenker et al. (1969) and Miehlich (1970) did not observe deterioration of soil physical properties underneath spruce but found increased macropore and total soil pore volumes in the organic mineral soil down to 15 cm soil depth. All other authors measured soil compaction at various soil depths, caused by clearcutting, the flat root system of spruce and its soil compacting effects, if wind-sway of the stems leads to vibration of the roots.

Hartge et al. (1983) studied changes of soil structure caused by older roots of *Prunus avium*. Radial growth of roots caused displacement of soil material. This led to compaction under thick tree roots (diameter > 5 cm), while the soil above them was lifted and loosened. Change in height of soil surface close to the trunk indicated lifting of soil material by radial growth, and partly by wind action on the tree. Hoglind and Nilsson (1989) found no differences between pure stands of Norway spruce and mixed species stands in porosity, dry bulk density and particle density in the upper 30 cm of the soil. This is in contrast to Daniluk et al. (1989), who measured increased water permeability along the roots of *Pinus sylvestris* and *Picea abies* on an average by a factor of 3.7, and of the soil within the root system by a factor of 1.2–1.9 due to wind induced vibrations of the roots and the ball of soil.

According to the citations above we can not generalize how Norway spruce changes soil physical properties in the upper or deeper mineral soil horizons. However, changes of soil physical properties, especially soil compaction, by species composition may affect reforestations of former pure spruce stands after clearcutting or gradual conversions towards mixed species stands. Machinery-induced soil compaction (e.g., Duval et al., 1989; Hakansson and Voorhees, 1997) and its negative impact on tree growth (e.g., Corns, 1988) is well documented. Korotaev (1992) measured root growth of 3-year plants of six species in differently compacted substrate and concluded that root growth was significantly negatively correlated with soil density. *Picea abies* was least capable of penetrating compacted soil, but the total root biomass produced was the same, as reduced root growth in the dense soil was compensated by more active root growth in the topsoil. In a pot experiment Hager and Sieghardt (1981) studied the impact of compacted soil on the growth of 4-year-old Norway spruces planted in soils with high and low content of coarse fractions. Dry mass of needles, stems and roots were significantly lower in the compacted substrate and these effects were more pronounced in the soil with the low content of coarse fractions. Hildebrand (1983) studied the impact of soil compaction on germination and growth of *Fagus sylvatica*, which would be the dominant natural deciduous tree of the study area. The soil function as seedbed for beech was completely

inhibited and the fine root development was reduced for planted trees at bulk densities greater than 1.25 g cm^{-3} .

This study was performed to address the following questions: (1) are top soil physical properties of secondary pure Norway spruce stands and natural mixed species stands on comparable sites different? (2) If there are significant differences, how do different soil types and stand development stages affect these properties? (3) Which soil physical parameter is most useful to describe differences? (4) How are selected soil parameters related to each other.

2. Study sites and methods

2.1. Study sites

Within the special research program (SRP) 'Restoration of Forest Ecosystems' 30 pairs of secondary pure spruce stands (*Picea abies*) and adjacent mixed species stands on comparable sites were established for studying differences between these forests (Berger et al., 1998). The aim of the SRP is to develop management tools for forest restoration, mainly converting pure coniferous stands to mixed species

forests. Forty-eight study sites (24 pairs) with different stand development stages (mature stage, pole stage) were selected for this study on two different bedrock materials (Molasse, Flysch).

2.1.1. Study sites on Molasse

These study sites are located near Mattighofen ($13^{\circ}08'E$, $48^{\circ}07'N$), Upper Austria, in a forested landscape, called Kobernausserwald, at elevations between 510 and 720 m (Fig. 1). Long term means of annual precipitation (1200–1500 mm) and annual temperature ($6.2\text{--}7.5^{\circ}\text{C}$) were calculated from nearby weather stations for the Kobernausserwald by Bauer (1989). Parent material for soil formation are tertiary sediments (so-called 'Hausruck-Kobernausserwald' gravel), which consist mainly of quartz and other siliceous material. Because of this acid bedrock material with low rates of nutrient release, the dominant soil types are semi-podzols (intermediate soil type between cambisol and podzol) and podzols. Humus forms are moder and mor and the thickness of the forest litter layer is between 5 and 10 cm, indicating slow turnover and accumulation of nutrients. pH values (CaCl_2) of the upper mineral soil (0–5 cm soil depth) are between 2.6 and 3.4. The natural forest

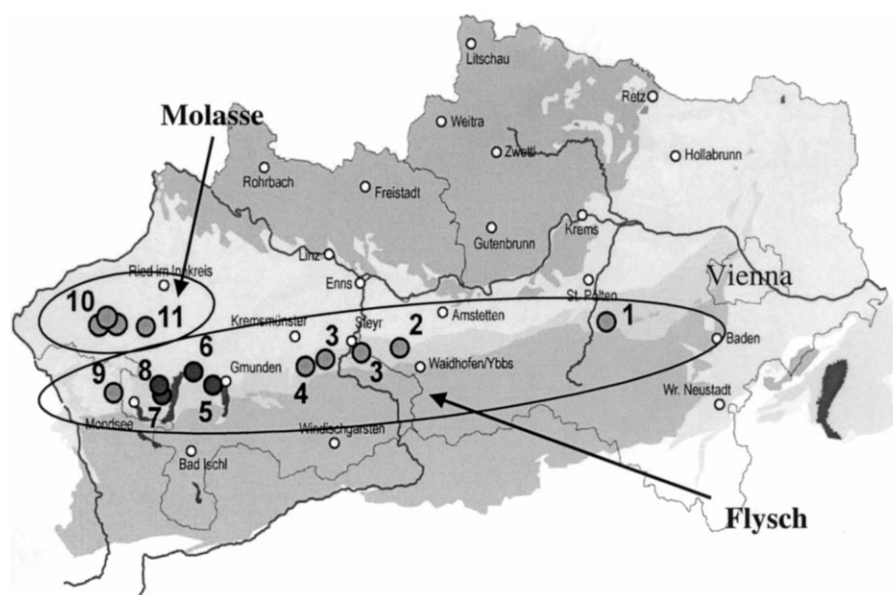


Fig. 1. Location of SRP study sites on Flysch and Molasse. Numbers indicate study areas of different land ownerships or management districts (modified from Hasenauer and Sterba, 1999).

vegetation of mixed species stands is *Luzulo nemorosae-Fagetum*.

2.1.2. Study sites on Flysch

The Flysch zone is a narrow strip in the foothills of the Northern Limestone Alps from west to east throughout the country. Hence, the study sites are spread throughout Lower and Upper Austria from approximately 13°22'E, 47°50'N (Mondsee) to 15°37'E, 48°11'N (St. Pölten) at elevations between 470 and 790 m (Fig. 1). Flysch consists mainly of old tertiary and mesozoic sandstones and clayey marls. Nutrient release from this bedrock material is higher in comparison to the Molasse and consequently the prevalent humus forms are moder and mull, indicating quick turnover of the forest litter layer (usually less than 5 cm thickness). pH values (CaCl₂) of the upper mineral soil (0–5 cm soil depth) are between 3.1 and 5.5. All soils of these study sites were classified as pseudogley (Schachtschabel et al., 1997; FAO classification: stagnic Gleysol), since horizons with a high fraction of fine material (loam, clay) cause temporary waterlogging. In general, soils on Flysch are less acidic, better supplied with nutrients, and less sandy than soils developed on Molasse. The natural forest vegetation of mixed species stands on Flysch soils is *Asperulo odoratae-Fagetum*.

2.2. Soil sampling

The litter layer was sampled with a sampling frame of 900 cm² area at approximately the same distances between two adjacent tree trunks (between trunks). After the litter layer was removed, two undisturbed mineral soil core samples (0–4 and 4–8 cm) were taken within the 900 cm² area. The core sampler was filled with two sample retaining cylinders (inner diameter: 5.4 cm and height: 4 cm) enabling separate sampling of 0–4 and 4–8 cm soil depth. There were five replications (each litter layer, 0–4 and 4–8 cm soil cores) at each site, randomly distributed. In addition, a second set of soil cores was taken within each frame area for measuring carbon of the fine soil fraction.

At the secondary pure spruce stands additional undisturbed soil core samples (five replications per site per depth) were taken from distances of $2.5 \times \text{DBH}$ (diameter in breast height, cm) downhill of a spruce trunk (trunk area). If the hypothesis is right,

that the flat rooting system of Norway spruce does cause soil compaction, it should be higher for the trunk area samples than for the between trunks samples, assuming that root pressure on the top soil is spread conically. Hence, soil compaction should be also detectable at approximately these spots, because no samples could be taken from the soil directly underneath coarse tree roots. The litter layer of the trunk area was removed, but not sampled.

2.3. Soil analysis

Samples from the forest litter layer were dried (105°C) for determining the dry mass of the forest floor (kg m⁻²) above the individual mineral soil core samples.

The retaining cylinders, containing the undisturbed soil cores, were sealed and transported to the laboratory, where the covers were taken off carefully and the bottom of the cylinder was sealed again with a polyethylene net (0.5 mm mesh width). The meshes of the net were small enough to prevent the soil from falling out of the cylinders, but big enough to enable water movement through it without considerable impact of capillary forces. The cylinders were put on another net (1 cm mesh width), placed 1.5 cm above the bottom of a box. Water was filled into the box until the water table reached the upper edge of the cylinders. Afterwards the box was closed and the samples were kept in water for 3 days. Suction was applied to the box periodically to take off air bubbles in the soil pores, which would prevent the soil from being completely saturated. Then, each cylinder was dipped two times for 3 s on a wet paper towel before the total weight of the cylinder was measured. Subtracting the masses of the cylinder and the net (0.5 mm mesh width) yielded the mass of the completely saturated soil core sample, which was used for further calculations.

Thereafter, each retaining cylinder, filled with the completely saturated soil core, was put back on the net (1 cm mesh width) of the box. Sealing the box and keeping a low water table below the net prevented the soil samples from losing water from evaporation. The samples were kept for 24 h in the box and weighed again.

The mass difference between completely saturated and 24 h drip-drained soil within the same cylinder was calculated for all five replications per site and the

median was determined, assuming that this sample is the stand's most representative one for further investigations. The net, which sealed the bottom of the cylinder was replaced by a ceramic plate and the water desorption at 10 kPa air pressure was measured for the chosen soil core according to Richards (1965) by means of a pressure plate apparatus (soil-moisture equipment).

Once the completely saturated soil cores were weighed after 24 h of free dripping and every fifth of them after water desorption at 10 kPa, the soil cores were taken out of the cylinders and the soil was separated into the fine soil (dry sieving, <2 mm) and the coarse fraction (wet sieving, >2 mm). Roots were sorted out by hand. Total dry masses of the fine soil, the coarse fraction and the roots were calculated after drying at 105°C.

All five replications per site and depth of the additional soil samples for chemical analyses were pooled and the fine soil (sieving, <2 mm) was analyzed for total content of carbon (LECO SC 444).

2.4. Calculations

The 24 h drip-drained macropore volume, used in this study, was defined as the loss of water after free dripping of completely saturated soils for 24 h in percent of the bulk volume.

The 10 kPa desorption macropore volume in percent of the total bulk volume was calculated from the loss of water after desorption at 10 kPa in a pressure plate apparatus (Richards, 1965). Water desorption at 10 kPa is mainly a function of soil structure and to some extent of soil texture. According to Schachtschabel et al. (1997), the change from primary (function of soil texture) to secondary pores (function of soil structure) occurs at an equivalent pore diameter > 50 µm (6 kPa). Hence, water desorption at 6 kPa would have been more appropriate for this study of impact of stand's composition on soil structure. However, the lowest pressure setting of the given pressure plate was 10 kPa. According to Friedrich (1992), fine roots may create secondary soil pores by incorporating pore diameters up to 10 µm, indicating that water desorption even above 10 kPa may still be partly a function of soil structure.

The remaining fine soil bulk density was calculated as the bulk density of the fine soil fraction only by

dividing the dry mass of fine soil (<2 mm) with the corresponding volume: volume of soil core minus volume of coarse fraction (>2 mm) minus volume of roots. Converting masses (measured) into volumes was done by assuming densities of 2.65 g cm⁻³ for the coarse fraction (Schachtschabel et al., 1997) and 0.40 g cm⁻³ for roots (compare Wagenführ, 1996 – beech: 0.51 g cm⁻³, spruce: 0.33 g cm⁻³). Note, that the remaining fine soil bulk density, used in this research, is not equal to the commonly used bulk density of the whole soil sample (called total bulk density in this study), including roots and coarse fractions (e.g., Blake, 1965). If the variability of the coarse fraction is high (i.e., high gravel fraction in soils developed on Molasse but almost no coarse fraction in soils on Flysch), it is hypothesized that the remaining fine soil bulk density is a more useful tool for detecting changes of soil structure caused by compaction (e.g., flat rooting system of *Picea abies*).

3. Results and discussion

3.1. Soil parameters of the whole dataset (five subsamples per site)

Means of total bulk densities, remaining fine soil bulk densities, dry masses of coarse fragments, roots and forest floor as well as carbon contents are listed in Table 1. A 2 × 2 × 2 ANOVA was performed for each soil parameter to test differences between the means of the three grouping variables: bedrock material (Flysch, Molasse), stand development stages (mature stage, pole stage) and species composition (pure spruce stand, mixed species stand). The observations used in these analyses were the means of five subsamples per forest site, collected at locations between tree trunks.

Both total bulk density as well as remaining fine soil bulk density are significantly higher for soils developed on Flysch and for the mixed species stands (Table 1). The fact that pure spruce stands have a lower bulk density of the upper mineral soil is the most important result of this study and is in accordance with some of the cited literature above (e.g. Hartge et al., 1983; Schmidt-Vogt, 1986; Daniluk et al., 1989; Friedrich, 1992). Stand development stages did not affect the bulk density of the soil significantly, although there

Table 1

Soil parameters of undisturbed soil core samples, taken at approximately the same distances between two adjacent tree trunks (between trunks) of 48 forest sites at 0–4 cm and 4–8 cm soil depths^a

Soil parameter	Bedrock material (1)		Development stage (2)		Species composition (3)		Interaction
	Flysch	Molasse	Mature	Pole	Pure	Mixed	
Total bulk density (g cm ⁻³)							
0–4 cm	0.81	0.56***	0.68	0.74 n.s.	0.63	0.78***	n.s.
4–8 cm	1.01	0.87***	0.95	0.96 n.s.	0.88	1.02***	n.s.
Remaining fine soil bulk density (g cm ⁻³)							
0–4 cm	0.77	0.32***	0.57	0.61 n.s.	0.54	0.63*	n.s.
4–8 cm	0.97	0.50***	0.76	0.80 n.s.	0.73	0.82*	(1) × (3)**
Coarse fragments (mg per cm ³ bulk volume)							
0–4 cm	54.0	271.7***	139.1	152.6 n.s.	108.6	180.8*	(1) × (2)**
4–8 cm	66.9	449.2***	239.8	207.2 n.s.	193.9	258.5*	(1) × (3)**
Roots (mg per cm ³ bulk volume)							
0–4 cm	2.38	4.21*	2.80	3.61 n.s.	3.80	2.48*	n.s.
4–8 cm	2.23	3.16 n.s.	2.91	2.22 n.s.	2.59	2.65 n.s.	n.s.
Carbon (%)							
0–4 cm	5.21	17.40***	11.47	8.63*	11.22	9.35 n.s.	n.s.
4–8 cm	3.31	10.81***	6.93	5.73 n.s.	7.11	5.76 n.s.	n.s.
Forest floor (kg dry mass per m ²)							
	1.17	3.38***	2.58	1.41***	2.37	1.81*	(1) × (2)**

^a The observations used in these analyses were the means of five subsamples per forest site. A 2 × 2 × 2 ANOVA was performed to test differences between means of the grouping variables bedrock material (Flysch – basic soil, Molasse – acid soil), stand development stages (mature stage, pole stage) and species composition (pure spruce stand, mixed species stand). Significant interactions between grouping variables indicate that these variables can not be tested individually but affect the dependent variable jointly.

*** Significance of each factor is shown as ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; n.s.: $p > 0.10$.

is a trend that bulk densities are higher under the younger stands (pole stage). This trend becomes significant for the remaining fine soil bulk density (total bulk density: not significant), if the ANOVA is performed for the pure spruce stands only ($N = 24$), indicating that either the young root system of Norway spruce did compact the upper soil or the soil above the old root system of Norway spruce was loosened (0–4 cm, mature stage: 0.50 g cm⁻³, pole stage: 0.59 g cm⁻³, $p < 0.05$; 4–8 cm, mature stage: 0.70 g cm⁻³, pole stage: 0.77 g cm⁻³, $p < 0.10$). Because total bulk densities take the coarse fraction into consideration, these values are higher than the corresponding remaining fine soil bulk densities. The observed differences in percent of total bulk densities are within a small range (4–5%) for soils on Flysch but amount up to 75% (74–75%) for soils on Molasse. These results are strengthened by the fact, that no significant interactions between the grouping variables were detected ($p < 0.05$), except for the remain-

ing fine soil bulk density at 4–8 cm soil depth between bedrock material and species composition. This kind of interaction was measured for other soil parameters as well and is interpreted that the impact of species composition on a given soil parameter depends on the soil type. In this case, the observed differences of bulk densities between species composition were higher for soils on Flysch than on Molasse (compare Table 2).

Higher amounts of coarse fragments in the soil on Molasse are in accordance with observed differences between total bulk density and remaining fine soil bulk density (Table 1). However, it is surprising that the upper mineral soil in mixed species stands has a higher coarse fraction than the soil in the pure spruce stand. According to Berger et al. (1998) the coarse fraction of the total profile (0–50 cm soil depth) revealed no differences between pure spruce and mixed species stands. Hence, we suggest that this difference in the upper mineral soil is caused by the species composition and not by inherent variation in soils within the

Table 2

Mean bulk densities (g cm^{-3}) of the total soil sample (tot.) and of the remaining fine soil fraction (r.f.s.)^a

Bedrock material	Development stage	Species composition	Mean bulk density (g cm ⁻³)		SD site means		N sites	SD sub-samples		N sub-samples	Level sign. ANOVA	
			tot.	r.f.s.	tot.	r.f.s.		tot.	r.f.s.		tot.	r.f.s.
0–4 cm soil depth												
Flysch	mature	pure	0.70	0.66	0.10	0.08	8	0.15	0.14	40	*	(*)
		mixed	0.90	0.85	0.11	0.12	8	0.17	0.17	40	*	**
	pole	pure	0.80	0.78	0.06	0.06	6	0.15	0.15	30	n.s.	n.s.
		mixed	0.84	0.82	0.19	0.20	6	0.21	0.22	30	***	***
Molasse	mature	pure	0.46	0.30	0.13	0.09	6	0.17	0.12	30	*	**
		mixed	0.58	0.34	0.13	0.07	6	0.21	0.11	30	(*)	(*)
	pole	pure	0.51	0.32	0.08	0.04	4	0.18	0.10	19	n.s.	n.s.
		mixed	0.73	0.35	0.17	0.08	4	0.19	0.09	20	**	**
4–8 cm soil depth												
Flysch	mature	pure	0.90	0.87	0.08	0.06	8	0.12	0.11	40	**	(*)
		mixed	1.14	1.08	0.07	0.11	8	0.10	0.14	40	**	**
	pole	pure	0.94	0.92	0.11	0.10	6	0.15	0.14	30	**	*
		mixed	1.05	1.03	0.11	0.14	6	0.15	0.17	30	**	***
Molasse	mature	pure	0.79	0.47	0.09	0.10	6	0.18	0.13	30	n.s.	**
		mixed	0.91	0.50	0.03	0.10	6	0.19	0.13	30	n.s.	**
	pole	pure	0.88	0.55	0.10	0.08	4	0.16	0.12	19	(*)	n.s.
		mixed	0.91	0.51	0.09	0.04	4	0.17	0.08	20	n.s.	n.s.

^a The remaining fine soil fraction was calculated by dividing the dry mass of fine soil (<2 mm) with the corresponding volume: volume of soil core minus volume of coarse fraction (density = 2.65 g cm^{-3}) minus volume of roots (density = 0.4 g cm^{-3}). Standard deviations (SD) are both listed for the site means (*N* sites = number of sites) and for all subsamples within a group (five subsamples per site, *N* subsamples = number of all subsamples). Grouping variables are soil depth, bedrock material, stand development stage and species composition. A one-way ANOVA (dependent variable: bulk density, factor: site) was performed to examine, whether the used statistical design enables detection of differences between sites within the same group.

*** Level of significance is shown as ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; (*): $0.05 < p < 0.10$; n.s.: $p > 0.10$.

selected pairs of forest stands. One explanation might be that by lifting and loosening (compare lower bulk densities) of the soil above the flat root system of Norway spruce the coarse fraction remains in deeper horizons. Another possible explanation would support the theory that due to higher amounts of forest floor under pure spruce the dividing line between mineral soil and the litter layer is rising without increase in the coarse fraction of the soil.

There are more living roots in the upper soil (0–4 cm soil depth) on Molasse, probably because lower pH, higher release of nutrients in the thicker litter layer and higher masses of the coarse fraction cause the root system to be more flat than in the soil on Flysch (Table 1). The fact that more roots were measured underneath the pure spruce stand than underneath the mixed species stand at depths from 0–4 cm is in accordance with the described flat root system of Norway spruce (e.g., Rottmann, 1989).

Each of the three factors had a significant impact on the dry mass of forest floor. Dry masses were higher for soils on Molasse, underneath mature stage and underneath pure spruce (Table 1). Consequently the carbon contents of the mineral soil showed a similar trend (significantly higher: Molasse, both depths, $p < 0.001$; mature stage, 0–4 cm, $p < 0.05$; pure spruce stand, 4–8 cm; $p < 0.10$).

Soil depth (0–4 and 4–8 cm) was tested in a separate $2 \times 2 \times 2$ ANOVA (bedrock material, species composition, soil depth) excluding the grouping variable stand development stage, which was not significant in most cases. Total bulk density, remaining fine soil bulk density and coarse fragments increased and carbon content decreased significantly ($p < 0.001$) from 0–4 to 4–8 cm soil depth. Soil depth did not affect the root distribution significantly for all stands. However, dry masses of roots declined from 0–4 to 4–8 cm soil depth within the pure spruce stands ($p < 0.10$), while

no decline was measured within the mixed species stands only. This is in accordance with Osenberg (1998), who found a maximum of root distribution above 10 cm soil depth for a pure spruce stand and in 15–20 cm soil depth for a mixed species stand on comparable sites.

No significant differences at all were detected between the locations ‘between trunks’ and ‘trunk area’ (not shown) within the pure spruce stands ($n = 24$) for total bulk density, remaining fine soil bulk density, coarse fragments and roots (neither at 0–4 cm nor at 4–8 cm soil depth) in another $2 \times 2 \times 2$ ANOVA (bedrock material, stand development stage, location).

3.1.1. Spatial variation

Mean total bulk densities and mean remaining fine soil bulk densities are given in Table 2 for all 16 groups, defined by the grouping variables soil depth, bedrock material, stand development stage and species composition. In all cases, both bulk densities are higher for the mixed species stand. The highest densities were measured in soils on Flysch at 4–8 cm soil depth under mature, mixed species stands (total bulk density: 1.14 g cm^{-3} , remaining fine soil bulk density: 1.08 g cm^{-3}) and the lowest densities were recorded in soils on Molasse at 0–4 cm soil depth under pure spruce stands in the mature stage (total bulk density: 0.46 g cm^{-3} , remaining fine soil bulk density: 0.30 g cm^{-3}).

Because bulk density is the best of our indicators of compaction, and was measured on all subsamples, we used variation among samples within sites as a measure of soil spatial heterogeneity. A one-way ANOVA (dependent variable: bulk density, factor: site) was performed to examine whether there were significant differences between sites within the same group. Five replication samples per site were enough to separate between sites in most cases (Table 2). At 0–4 cm soil depth the model was significant for all cases except for pure spruce stands in the pole stage on both bedrock materials. On Flysch, at 4–8 cm soil depth, variations per site were small enough to differentiate between sites within all groups. However, on Molasse (4–8 cm soil depth) the used statistical design did not differentiate between sites for total bulk density ($p < 0.05$), while it did for the remaining fine soil bulk density under mature stands. This fact supports the hypothesis

(see section calculations) that the remaining fine soil bulk density is more useful than the total bulk density for soils with high variations of the coarse fragments.

3.1.2. Correlations between selected soil parameters

Correlation analyses were performed on data of the individual soil core samples collected between the trunks ($N = 239$ per soil depth). A matrix of the correlation coefficients between all analyzed soil parameters, including the 24 h drip-drained macropore volume, is listed in Table 3. For the regressions between carbon and the other parameters, site averages had to be used, because five replicate samples per site were blended before carbon analysis ($N = 48$ per soil depth).

The remaining fine soil bulk density was negatively correlated (for all cases $p < 0.001$, except $p < 0.10$ with roots at 4–8 cm soil depth) with all other soil parameters (coarse fragments, roots, carbon, dry mass of forest floor, 24 h drip-drained macropore volume). Again, total bulk density was less useful as indicator of soil physical properties, indicated by constantly lower (partly not significant, not shown) correlation coefficients. However, high correlation coefficients between the remaining fine soil bulk density (this study) and the commonly used (total) bulk density (0–4 cm: $r = 0.83$, $p < 0.001$; 4–8 cm: $r = 0.66$, $p < 0.001$) still support the use of (total) bulk density as measure of soil compactness. According to Hakansson and Voorhees (1998) (total) bulk density is useful as long as only one soil is regarded, but it is no longer a suitable parameter for characterization of the state of compactness, when comparing across different soils.

Coarse fragments, roots, carbon, forest floor and 24 h drip-drained macropore volume (all analyzed parameter except bulk density) were positively correlated with each other. Although the organic carbon content is a chemical soil parameter it strongly effects soil physical parameters. High carbon content in the soil reduces the remaining fine soil bulk density significantly (correlation coefficients are between -0.83 and -0.87 , $p < 0.001$). The 24 h drip-drained macropore volume is positively correlated with the soil carbon content, because high carbon content causes the formation of very porous soil aggregates (Schmidt-Vogt, 1986). Hence, lifting and loosening of the soil above the root system (e.g., by radial growth and wind induced vibrations of the roots) is probably

Table 3
Pearson correlation coefficients (*R*) between selected soil parameters^a

Coarse fragments	Roots	Carbon	Forest floor	Macropore volume (24 h drip-drained)	
0–4 cm soil depth					
–0.47***	–0.27***	–0.83***	–0.63***	–0.29***	remaining fine soil bulk density (g cm ^{–3})
–	0.01 n.s.	0.48***	0.29***	0.19**	coarse fragments (mg per cm ³ bulk volume)
	–	0.34*	0.17*	0.00 n.s.	roots (mg per cm ³ bulk volume)
		–	0.78***	0.22**	carbon (%)
			–	0.15*	forest floor (kg dry mass per m ²)
				–	24 h drip-drained macropore volume (%)
4–8 cm soil depth					
–0.71***	–0.12 (*)	–0.87***	–0.66***	–0.52***	remaining fine soil bulk density (g cm ^{–3})
–	0.00 n.s.	0.79***	0.42***	0.49***	coarse fragments (mg per cm ³ bulk volume)
	–	0.19 n.s.	0.19**	–0.08 n.s.	roots (mg per cm ³ bulk volume)
		–	0.78***	0.60***	carbon (%)
			–	0.41***	forest floor (kg dry mass per m ²)
				–	24 h drip-drained macropore volume (%)

^a Regressions were performed on data of individual soil core samples (five subsamples per plots, between trunks area, *N* = 239), except for regressions between carbon and the other parameters, site means were used (*N* = 48). The 24 h drip-drained macropore volume represents the loss of water after free dripping of completely saturated samples for 24 h, expressed in percent of total bulk volume.

*** Level of significance is shown as ***: *p* < 0.001; **: *p* < 0.01; *: *p* < 0.05; (*): 0.05 < *p* < 0.10; n.s.: *p* > 0.10.

not the only reason for the observed lower bulk density underneath pure spruce stands. Higher carbon content in the top soil under pure spruce stands than under mixed species stands (see Table 1) and the fact that carbon may increase the macropore volume of the soil (Table 3) could be another reason for the observed decrease in soil bulk density under Norway spruce.

3.2. Soil parameters of the selected subset (one sample per site)

As described earlier, the mass difference between completely saturated and 24 h drip-drained soil was measured for all five replications per site and the median was selected for measuring the 10 kPa water desorption, assuming that this sample is the stand's most representative one.

A 2 × 2 × 2 ANOVA (bedrock material, stand development stage, species composition) for the between trunks samples was not significant for 0–4 cm soil depth but revealed significant lower 24 h drip-drained macropore volumes for soils at 4–8 cm depth on Flysch and for the younger stands (pole stage). Because the factor location did not differentiate between the samples, the same model was performed

on the larger dataset ('between trunks' and 'trunk area'), indicating significantly lower values for soils on Flysch than on Molasse at 0–4 cm soil depth as well. As will be discussed later, measuring the 24 h drip-drained macropore volume by means of free drainage of saturated soils was problematic in regard to absolute values (therefore not given in the preceding chapter), but was a useful tool for relative comparisons between the soil core samples.

3.2.1. 10 kPa water desorption

Means of the 10 kPa desorption macropore volume are listed for the grouping variables bedrock material, stand development stage and species composition in Table 4. The same kind of ANOVA, which was performed for the soil parameters, based on the whole dataset (one mean of five replications per site, compare Table 1), was done for this parameter as well (one selected sample per site, Table 4). Obviously, the chosen sample was highly representative for the whole stand, because the results obtained for the 10 kPa desorption macropore volume are exactly in accordance with those obtained for the remaining fine soil bulk density. Hence, low remaining fine soil bulk density can be used as a complementary parameter

Table 4

10 kPa desorption macropore volume (%) of undisturbed soil core samples, taken at approximately the same distances between two adjacent tree trunks (between trunks) and at distances of $2.5 \times \text{DBH}$ (diameter in breath height, cm) downhill of a spruce trunk (trunk area) of 48 forest sites at 0–4 and 4–8 cm soil depths^a

Location	Bedrock material (1)		Development stage (2)		Species composition (3)		Interaction
	Flysch	Molasse	Mature	Pole	Pure	Mixed	
Between trunks (cm)							
0–4	20.1	27.8**	23.9	22.4 n.s.	25.9	20.7*	n.s.
4–8	15.1	22.8***	18.8	17.7 n.s.	20.7	16.0**	(1) $\times$ (3)**
Trunk area (cm)							
0–4	25.3	27.1 n.s.	29.0	21.9**	–	–	n.s.
4–8	18.4	22.4*	21.3	18.4 (*)	–	–	n.s.

^a The 24 h drip-drained macropore volume of 5 samples per site was recorded and the median was chosen for measuring the water desorption (10 kPa air pressure) according to Richards (1965). A $2 \times 2 \times 2$ ANOVA ($N = 48$) was performed for the between trunks samples to test differences between means of the grouping variables bedrock material (Flysch – basic soil, Molasse – acid soil), stand development stages (mature stage, pole stage) and species composition (pure spruce stand, mixed species stand). Because samples of the trunk area were only taken of the pure spruce stands a 2×2 ANOVA (without the grouping variable species composition, $N = 24$) was performed for these samples. Significant interactions between grouping variables indicate that these variables can not be tested individually but affect the dependent variable jointly.

*** Significance of each factor is shown as ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; (*): $0.05 < p < 0.10$; n.s.: $p > 0.10$.

for high 10 kPa desorption macropore volume. The 10 kPa desorption macropore volume was significantly lower for soils developed on Flysch and underneath mixed species stands (Table 4). Stand development stages did not affect the 10 kPa desorption macropore volume of the soil between trunks significantly, although there is a trend that volumes are higher under the mature stands. Performance of the ANOVA for the pure spruce stands only ($N = 24$) for the soil between trunks did not reveal significant differences in contrast to the remaining fine soil bulk

density, however this trend became significant for soil samples collected from the trunk area (see Table 4). As expected, the 10 kPa desorption macropore volume was higher ($p < 0.001$) in soil samples from 0–4 cm soil depth than from 4–8 cm soil depth, while no significant differences were recorded for both depths between the two locations ‘between trunks’ and ‘trunk area’.

Correlation coefficients between the 10 kPa desorption macropore volume and other soil parameters are given in Table 5. The described link between bulk

Table 5

Pearson correlation coefficients (R) between 10 kPa desorption macropore volume (%) and selected soil parameters ($N = 48$, between trunks area)^a

Soil parameter	0–4 cm soil depth	4–8 cm soil depth
Total bulk density (g cm^{-3})	–0.42**	–0.62***
Remaining fine soil bulk density (g cm^{-3})	–0.71***	–0.78***
Coarse fragments (mg per cm^3 bulk volume)	0.15 n.s.	0.56***
Roots (mg per cm^3 bulk volume)	0.37*	0.19 n.s.
Forest floor (kg dry mass per m^2)	0.37**	0.52***
24 h drip-drained macropore volume (%)	0.48**	0.52***
Total pore volume (%)	0.68***	0.59***

^a The 24 h drip-drained macropore volume represents the loss of water after free dripping of completely saturated samples for 24 h, while the 10 kPa desorption macropore volume was measured after pressure plate extraction at 10 kPa (Richards, 1965). Total pore volume was calculated assuming a density of 2.65 g cm^{-3} for solid soil particles (fine soil, coarse fraction) and 0.4 g cm^{-3} for roots.

*** Level of significance is shown as ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; n.s.: $p > 0.10$.

density and 10 kPa desorption macropore volume is expressed by significant negative coefficients, whereby once more the remaining fine soil bulk density ($R = -0.71$ to -0.78) is more useful for characterizing the state of compactness than the total bulk density ($R = -0.42$ to -0.62). Mass of coarse fragments (except at 0–4 cm soil depth), mass of roots (except at 4–8 cm soil depth), dry mass of forest floor, 24 h drip-drained macropore volume and total pore volume are all positively correlated with the 10 kPa desorption macropore volume.

3.2.2. Soil pore volumes

The 24 h drip-drained macropore volume, the 10 kPa desorption macropore volume and the rest of the total soil pore volume (>10 kPa) are plotted in Fig. 2 for all 16 groups (between trunks), defined by the grouping variables soil depth, bedrock material, stand development stage and species composition. The total length of the columns represents the total soil pore volume, which was calculated by means of measured dry masses of the coarse fraction, fine soil

and roots and the assumed corresponding densities of 2.65 g cm^{-3} (coarse fraction, fine soil) and 0.40 g cm^{-3} (roots), respectively.

For each soil depth, total soil pore volume was significantly lower for soils on Flysch (0–4 cm: 69%, 4–8 cm: 61%) than on Molasse (0–4 cm: 76%, 4–8 cm: 64%) and for samples underneath the mixed species stands (0–4 cm: 69%, 4–8 cm: 60%) than underneath the pure spruce stands (0–4 cm: 74%, 4–8 cm: 65%). The assumed density of 2.65 g cm^{-3} for the fine soil fraction represents a maximum value, because high carbon contents cause lower densities of the fine soil fraction in comparison to the solid bedrock material. Carbon contents were not analyzed for the individual soil cores for which the water desorption at 10 kPa was measured, but this possible error of the total soil pore volume was estimated: using the mean carbon content (five replications per site) and a factor for reducing the fine soil density according to Katzensteiner (1992; fine soil density (in g cm^{-3}) = $2.65 - 0.02327 \times C$ (in %)) reveals that the given mean total soil pore volumes of all groups

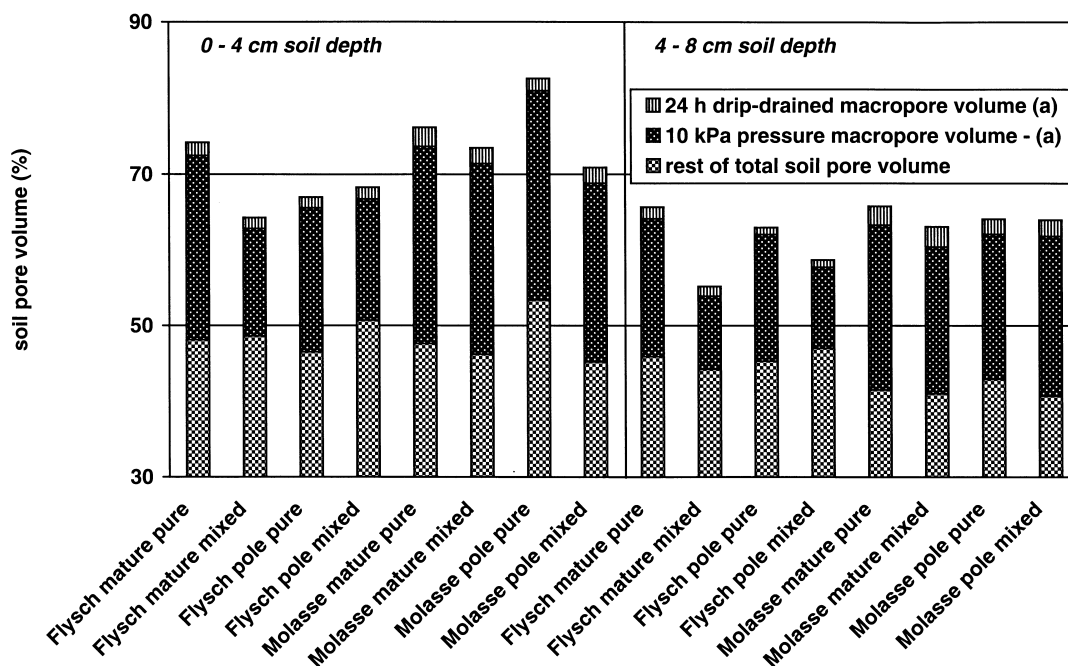


Fig. 2. Means of total soil pore volume (whole bar length), 24 h drip-drained macropore volume and 10 kPa desorption macropore volume (reduced by 24 h drip-drained macropore volume) in % bulk volume of the grouping variables bedrock material (Flysch, Molasse), stand development stages (mature stage, pole stage) and species composition (pure spruce stand, mixed species stand) for 0–4 and 4–8 cm soil depth.

is overestimated relatively to the corrected total soil pore volume by only 1.4–3.1%. Hence, it is concluded that errors by assuming the same density for coarse and fine fractions for all samples are easily within the range of sampling, handling and analytical errors and do not alter substantially relative differences between the groups.

Total soil pore volume was also estimated by subtracting the dry mass of the whole soil sample from the mass of the completely saturated sample. By doing so, the calculated total soil pore volume was lower (relative differences were 11–24% (0–4 cm) and 9–16% (4–8 cm)) than the corresponding volume as calculated from means of soil densities as described above. This difference is probably due to difficulties in complete saturation of upper mineral soil samples, which are rich in hydrophobic organic matter. Another reason might be the fact that the saturated samples were dipped two times for 3 s on a wet paper towel before the weight was measured to get rid of drops at the outside of the retaining cylinder and the net, unfortunately partly draining big macropores as well. Hence, it is concluded that the 24 h drip-drained macropore volume (1–4%) was underestimated in this study, because part of that difference should have been added to it. However, the 24 h drip-drained macropore volume was useful in this study for relative comparisons between the soil core samples and the selection procedure as described earlier.

The rest of the total soil pore volume in Fig. 2 (water desorption > 10 kPa) is not a function of soil structure but of soil texture. Despite the differences of the two bedrock materials in soil texture no differences were found for the upper 0–4 cm soil depth. At 4–8 cm soil depth the pore volume (>10 kPa) of the fine-textured (loam, clay) soil on Flysch (46%) was significantly higher than on the coarse-textured (sand, loam) soil on Molasse (41%).

4. Conclusions

It was hypothesized that soil physical properties of secondary pure Norway spruce stands and natural mixed species stands on comparable sites are different. Because soil compaction caused by the flat root system of spruce may have a negative impact on the soil as seedbed and for seedling establishment as well

as on the growth of trees this topic is important for forest restoration, especially when secondary pure spruce stands are to be converted to mixed species stands.

The remaining fine soil bulk density was a useful parameter for characterization of the state of compactness. Pure spruce stands caused a lower bulk density of the upper mineral soil due to lifting and loosening of the soil above the root system (compare Question 1, Section 1). Although the conclusions of the available literature are contradictory about soil compacting effects due to wind induced vibrations of the roots, the results of this study are in accordance with most of the cited literature if they are seen in more details according to Friedrich (1992), who concludes that the root system of spruce loosens the upper mineral soil while the lower mineral soil (below 30–40 cm) is compacted.

In some cases an ANOVA revealed significant interactions between the grouping variable bedrock material and species composition, indicating that the impact of species composition on a given soil parameter depends on the soil type (Question 2). Hence, these variables can not be tested individually but affect the dependent variable jointly. Because soils on Molasse (semi-podzols and podzols) were characterized by low pHs and high coarse fractions, dry masses of forest floor and living roots as well as carbon contents in the upper most mineral soil horizons were higher than for soils developed on Flysch (pseudogleys; less acidic, high fraction of fine material). Bulk densities were lower for soils on Molasse.

The remaining fine soil bulk density was the most useful parameter for characterizing the impact of the species composition, taking into consideration that it is easy to determine (Question 3). Results of calculated macropore volumes after water desorption at 10 kPa pressure plate extraction were exactly conform with those obtained for the remaining fine soil bulk density, indicating significant differences for the grouping variables bedrock material and species composition. Total bulk density was not as useful as the remaining fine soil bulk density for separating between the different grouping variables due to high variations of the coarse fragments in soils on Molasse. The 24 h drip-drained macropore volume was useful for relative comparisons between the soil core samples, but did not provide correct absolute values.

The remaining fine soil bulk density was negatively correlated with all other soil parameters (coarse fragments, roots, carbon, dry mass of forest floor, 24 h drip-drained macropore volume). Because low remaining fine soil bulk density can be used as a complementary parameter for high 10 kPa desorption macropore volume, the same observed significant correlations were positive between the 10 kPa desorption volume and those parameters. Coarse fragments, roots, carbon, forest floor and 24 h drip-drained macropore volume were positively correlated with each other (Question 4).

It could be concluded from this study that changes of soil physical properties of the upper mineral soil (0–8 cm soil depth) by Norway spruce may not reduce germination and establishment of mixed species trees. However, chemical and nutritional changes were not subject of this study, which are expected to limit the success of forest restoration, e.g., conversion of secondary pure spruce stands towards natural mixed species stands.

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